

Dr. Hariharan Srinivasan

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Research Objective & Interest

- Objective** ■ To advance the understanding of complex dynamical systems by investigating the mechanisms underlying natural phenomena and their prediction using machine learning. I aim to develop robust frameworks that not only explain the emergence of such phenomena but also provide tools for anticipating and controlling them in real-world systems.
- Interests** ■ Extreme events, Stochastic systems, Machine learning (Developing new algorithms), Neuronal modeling, Predicting complex datasets, Model-Data integration, Fractional-Order systems and Solar physics.

Education

- 2022 – 2025 ■ **Ph.D., SASTRA Deemed to be University**, Thanjavur, TamilNadu, India.
- 2019 – 2021 ■ **M.Sc. Physics, SASTRA Deemed to be University**, Thanjavur, TamilNadu, India.
CGPA: 8.6829/10
- 2016 – 2019 ■ **B.Sc. Physics , SCSVMV University**, Kanchipuram, Tamil Nadu, India.
CGPA: 9.13/10

Skills

- Programming** ■ FORTRAN (Numerical computing), Python (Data analysis, Machine learning), MATLAB (Simulations, Plotting)
- Software/Tools** ■ LaTeX (scientific writing and typesetting), Mathematica (symbolic computation), GNUPlot, XPPAUT (Bifurcation analysis), MS Office
- Research Skills** ■ Scientific Writing, Manuscript Preparation, Data Analysis, Presentation
- Misc.** ■ Teaching, OS - Windows, Linux

Research Experience

Postdoctoral Research

- Present** ■ [Postdoctoral Researcher \(Senior Project Officer\)](#)
Indian Institute of Technology - Madras (IITM), Chennai, India
FedEx Industry Sponsored Project

Research Experience (continued)

Doctoral Research

- Ph.D.  Centre for Nonlinear Science & Engineering, Department of Physics, School of EEE, SASTRA Deemed to be University.
Thesis Title: **Exploring Noise-Induced Extreme Events in Neuronal Oscillators, Networks and Machine-Learning Forecasts**
Advisor: Dr. R. Suresh
- Explored the under-explored phenomenon of **noise-induced extreme events (EE)** in single neuronal models and deciphered the mechanisms using analytical and numerical techniques
 - Investigated the effects of heterogeneous noise in inducing EE in globally coupled and multiplex networks of neuronal oscillators.
 - Introduced a novel measure to characterize EE in microscopic and macroscopic scale
 - Developed a **novel echo state network (ESN)** algorithm called **clustered ESN** particularly to predict EE in multivariate time series data and explored the potential of various network topologies.
 - Applied the developed algorithm to predict complex datasets: Stock Market & Solar wind data.
 - Published in peer-reviewed journals and present my work in conferences.

Postgraduate research project

- M.Sc  Thesis title: **Advent of Extreme events in the Sun.**
Advisor: Dr. R. Suresh
- Applied **Detrended Fluctuation Analysis (DFA)** to sunspot data to investigate the presence of long-memory effects.
 - Isolated extreme data and analyzed them using DFA to understand their long-range dependence properties.
 - Results indicated that extreme events in sunspot activity are long-range correlated.
 - Performed additional characterizations including long-tail distribution and inter-event interval (IEI) distribution, confirming the presence of extreme events.

Research Publications

Journal Articles

- 1 **Hariharan, S.,** Suresh, R., & Chandrasekar, V. (2026a). Heterogeneous noise-induced extreme events and synchronization in a globally coupled network of fitzhugh–nagumo oscillators. *Chaos, Solitons & Fractals*, 203, 117652.  <https://doi.org/10.1016/j.chaos.2025.117652>
- 2 **Hariharan, S.,** Suresh, R., & Chandrasekar, V. (2026b). Synchronization and extreme events in multiplex fitzhugh–nagumo networks under heterogeneous noise. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 36(3).  <https://doi.org/10.1063/5.0304868>
- 3 **Hariharan, S.,** Suresh, R., & Chandrasekar, V. K. (2025). Noise-induced extreme events in single fitzhugh–nagumo oscillator. *Chaos, Solitons & Fractals*, 192, 116077.  <https://doi.org/10.1016/j.chaos.2025.116077>
- 4 **Hariharan, S.,** Suresh, R., & Chandrasekar, V. (2025). Multivariate time series prediction using clustered echo state network. *The European Physical Journal Plus*, 140(11), 1133.  <https://doi.org/10.1140/epjp/s13360-025-07077-3>
- 5 **Hariharan, S.,** Suresh, R., & Chandrasekar, V. K. (2024). Noise-induced extreme events in integer and fractional-order memristive hindmarsh–rose neuron models: A comprehensive study. *The European Physical Journal Plus*, 139(3), 292.  <https://doi.org/10.1140/epjp/s13360-024-05059-5>

Conference Presentations

Ph.D

- **Unveiling Noise-Induced Extreme Events: From Single Neuron to Network Dynamics**
Oral presentation at *Conference on Nonlinear Systems and Dynamics (CNSD-2025)*, Bharathidasan University, Trichy, India, 10–13 March 2025.
- **Noise Induced Extreme Events in Fractional Order Memristor based Hindmarsh–Rose Neuron Model**
Poster presented at *Complexity and Nonlinear Dynamics in STEM (CNLDS-2023)*, IIT Hyderabad, India, 5–7 June 2023.

M.Sc

- **Long memory in the extreme events of the Sun**
Poster presented at *13th Conference on Nonlinear Systems and Dynamics (CNSD-2021)*, SASTRA University, Thanjavur, India, December 2021.

Roles Undertaken During Ph.D

- | | |
|--------------------|---|
| Project Assistant | ■ Aug. 2023- Dec. 2025 |
| | Worked in the project titled “Investigation of emergence of extreme events due to heavy wind loads in medium and high rise building structures” sanctioned by DST - SERB, India along with my thesis. |
| Teaching Assistant | ■ Apr. 2022 - July. 2023 |
| | Handled M.Sc(Physics), B.Sc (Physics) and B.Tech physics laboratories. |

Certificate courses

- **Online Short Course on Solar Physics**
Organized by the *Centre for Space Science and Technology Education in Asia and the Pacific (CSSTEAP)* and conducted by the *Physical Research Laboratory, Ahmedabad*.
- **Quantum Mechanics for Scientists and Engineers**
Online course offered by *Stanford University, London*.

References

- 1 **Dr. R. Suresh**
Assistant Professor -II,
Centre for Nonlinear Science & Engineering,
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- 2 **Prof. V. K. Chandrasekar**
Associate Professor,
Centre for Nonlinear Science & Engineering,
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References (continued)

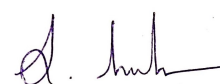
- 3 **Dr. R. Gopal**
Assistant Professor -II,
Centre for Nonlinear Science & Engineering,
Department of Physics, School of EEE,
SASTRA Deemed to be University,
Thanjavur, TN, India
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Personal details

Father's Name: Dr. V. Srinivasan
Date of Birth: 15/10/1998
Age: 27
Gender: Male
Nationality: Indian
Languages Known: English (Read, Write and Speak), Tamil (Read, write and speak), Telugu (Speak)
Marital Status: Unmarried
Address: No.6, Aswin Emerald, Block-A, AF-1, Periyar Nagar, 3rd Street, Madipakkam, Chennai,
TamilNadu, India - 60091

Declaration

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.



Hariharan S